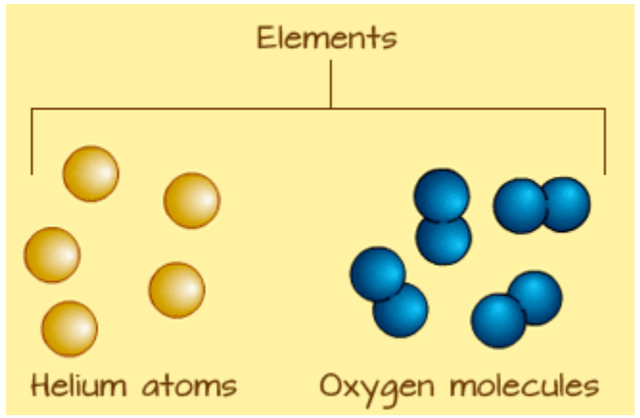
Knowledge Organiser
Year 7: Chemicals – Atoms, Elements and Compounds

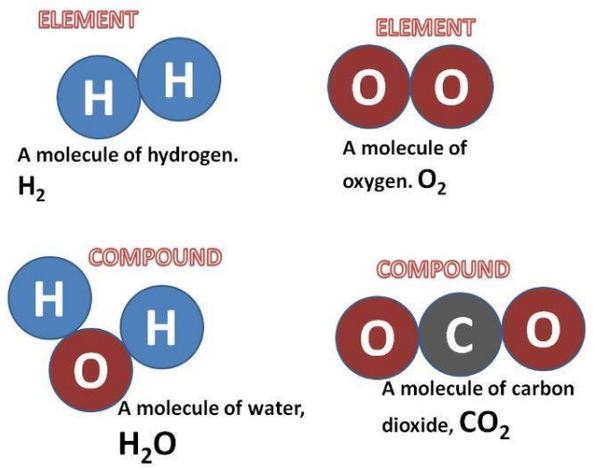
What is an element?
- Atoms are the smallest units of matter
- Elements are unique types of atoms
- A sample of an element is made from only one atom
- Each element has unique chemical and physical properties

A diagram showing the difference between atoms and elements

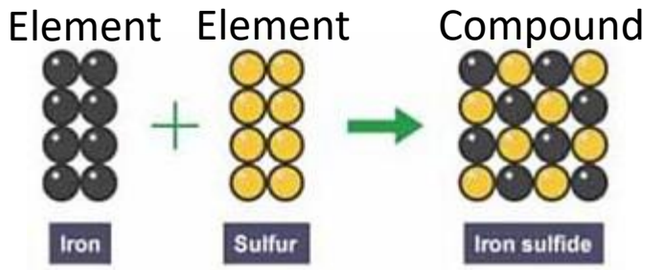


But what is a molecule?
- A molecule is when **more than one atom** bonds together
- They can be the same or different elements
- They bond in **fixed proportions** e.g. water molecules have 2 hydrogen and 1 oxygen atoms

A diagram showing the difference between molecules and compounds

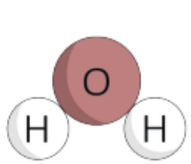


What is a compound?
- A chemical compound is formed when two or more elements are **chemically combined**
- The chemical compound will have different properties to the elements it made from
- Sodium chloride (salt) is a white **crystalline compound** formed from sodium, a silvery reactive metal and chlorine, a yellow green gas



How can we represent a chemical compound
- The atoms in a chemical compound are chemically bonded
- We can show this in lots of different ways
- A formula tells you how many atoms of each element there are
- CO_2 is the formula for carbon dioxide
- We can also use particle diagrams like the ones below

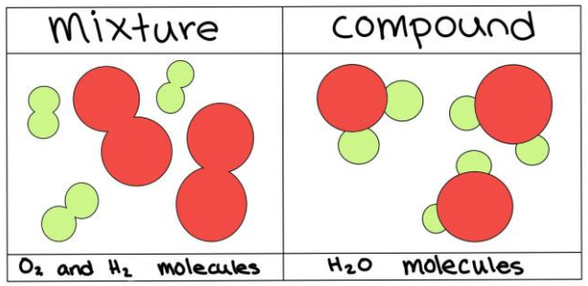
Bonds are shown with the atoms touching



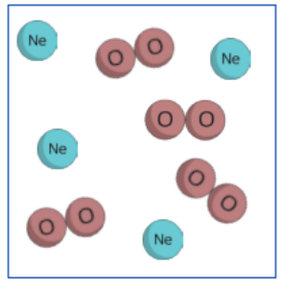


Bonds shown as sticks between the atoms

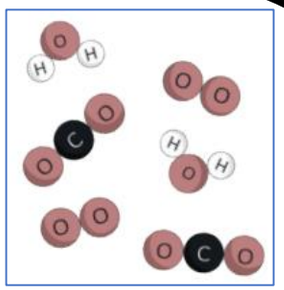
Separating chemical compounds?
- Chemical compounds are formed when atoms of different elements **chemically bond**
- These new bonds are very strong
- A mixture of iron and oxygen gas could easily be separated using a magnet
- But it would be impossible to separate iron and oxygen atoms from each other in rust (iron oxide)



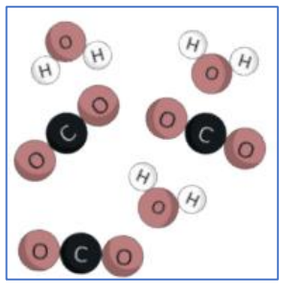
A diagram showing the difference between mixtures and compounds



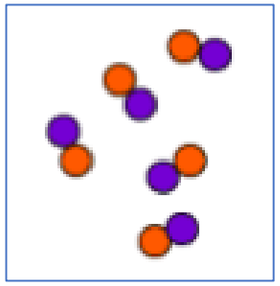
Mixture of elements



Mixture of compounds and elements



Mixture of compounds

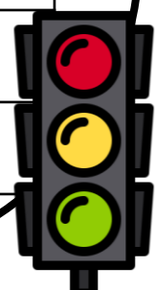


A pure sample of a compound

Key Term	Definition
Atom	Smallest unit of matter
Bond	An attraction between two atoms
Compound	Two or more elements bonded together
Element	Made from one type of atom
Mixture	Two or more substances not bonded together
Molecule	Two or more atoms bonded together

RAG Yourself!

Learning Outcomes;	RAG
Define atom and element	
Describe how a molecule is formed	
Describe how a chemical compound is formed	
Compare a mixture and a chemical compound	
Explain why the properties of a compound are different from the elements which make it	
Explain why the elements in chemical compound are difficult to separate	



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What is an element?

A diagram showing the difference between atoms and elements

What is a compound?

Element Element Compound

Mixture of elements

Mixture of compounds and elements

How can we represent a chemical compound

Mixture of compounds

A pure sample of a compound

Bonds are shown with the atoms touching

Bonds shown as sticks between the atoms

But what is a molecule?

A diagram showing the difference between molecules and compounds

Separating chemical compounds?

A diagram showing the difference between mixtures and compounds

Key Term	Definition
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Bond	
Compound	
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Mixture	
Molecule	

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